Experiment 5

Reflection and Refraction

Objective	Materials Used
Confirm the laws of Reflection and Refraction and examine the critical angle.	Raw Table and Accessories, Long sheet of white paper.

Background

When examining physical optics, it is important for students to have a basic understanding of how light interacts with different materials. The two basic kinds of interactions are reflection and refraction.

Law of Reflection: $\theta_i = \theta_r$
Law of Refraction: $n_1 sin(\theta_1) = n_2 sin(\theta_2)$

In each experiment, we will use a ray table to mark the angle of light entering or exiting the system. This allows for a light source to be placed on at the same location while changing the angle of interaction with the mirror or block of acrylic. The ray table will be placed on top of a long piece of white paper which is used to trace out the path of light as it interacts with the experiment. Make sure to think about how much space you need for tracing based on the setup you are using.

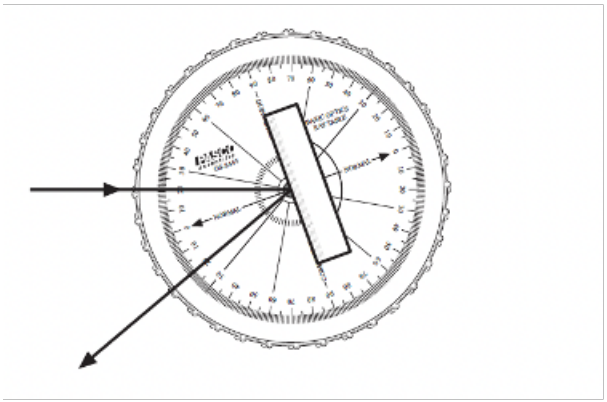
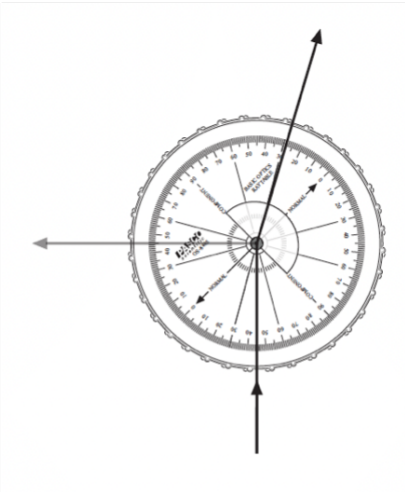


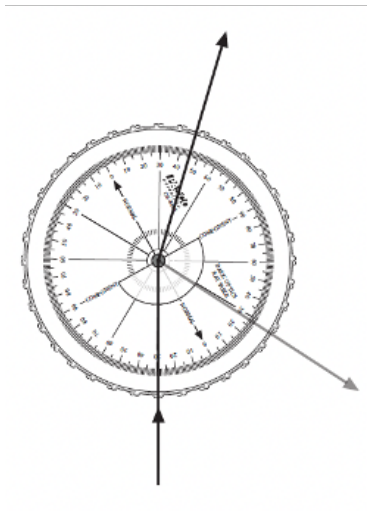
Figure 5.1: Ray Table (Mirror)

Procedure

1. Reflection: Plane Mirror
   (a) Set your light source such that only a single ray of light is emitted. Place the light source such that you can see the light on the ray table (Figure: 5.1). Put the flat mirror at the center of the ray table. Rotate the ray table such that the incident angle is 0 degrees.
   (b) Perform the reflection experiment by changing the incoming angle and measuring the outgoing angle. Do this for five angles between 10 and 80 degrees.
   (c) Compare the incoming and outgoing angles to confirm the law of reflection.
2. Reflection: Curved Mirror
   (a) Turn the light source to produce three parallel rays. Place the curved mirror (concave) on a sheet of paper along with the light source. Shine the rays on the curved mirror such that the central ray hits the mirror at the center of the curve. Trace the three reflected rays and note where they intersect *Do not move the mirror.*
   (b) Switch the light source back to the single light ray. Angle the light source such that the light reflects directly back along the path of the incoming ray. Do this for the center of the curve and the two extremes of the mirror. These reflections represent the radius of the circle used to create the mirror.
   (c) Confirm that the focal point of the curved mirror is half the radius of the circle.
3. Refraction
   (a) Return the light source to the ray table , and place the curved acrylic at the center of the ray table. Make sure that when the light hits the half circle, it hits the flat side at the exact center and then moves through the curved side as shown in figure 5.2a
   (b) Turn on the light and confirm you can measure the incoming and outgoing rays. Rotate the table such that the incident angle is 0 degrees.
   (c) Measure the refracted angle based on the incoming angle. The incoming angle should be every 5 degrees between 10 and 60 degrees.
   (d) Calculate the sine of the incident and refracted angle. Construct a plot to measure the index of refraction based on the Law of Refraction.
   (e) Another way to measure the index of refraction is through the critical angle. This is when all the light is reflected, and no light is refracted through the material. However, this only works if the light travels from a high index of refraction to a low index of refraction. Turn around the piece of acrylic to as show in figure 5.2b.
   (f) Find the angle at which the refracted angle becomes 90 degrees. This should be done by testing different angles.
   (g) Based on the Law of Refraction and your critical angle, calculate the index of refraction of the material. Compare this with the value measured from your plot.



(a) Ray Table (Acrylic) 1



(b) Ray Table (Acrylic) 2

Figure 5.2: Ray Tables for Refraction

4. Dispersion
   (a) Turn your light source to the setting with three rays of different colors. Use a sheet of paper to trace the incoming and outgoing rays.
   (b) Following one color at a time, send the light through your piece of acrylic as done in Procedure 3 step c for an incoming angle of 60 degrees. Confirm it is the same incident angle for all colors.
   (c) Mark the path of each color of light and see if you can measure a difference in the paths. Based on this, calculate the index of refraction for the most extreme paths. Note how these differ from your previously measured index of refraction.
5. Polarization
   (a) Use the microwave emitter and transmitter to examine polarization. Setup the transmitter and receiver as shown in figure 5.3.
   (b) There are two ways to measure the relative intensity of the wave. Use either the multimeter to measure voltage or the built in dial.
   (c) Adjust the settings on the receiver such that the relative signal is around 1 for the dial on the transmitter.
   (d) Using the knob on the back, rotate the transmitter and record the relative signal based on this angle. Record this for angles 0 to 180 degrees in steps of 10 degrees.
   (e) Plot the signal vs angle based on your measurements.
   (f) If the measurement is proportional to the electric field, this will be represented as a cosine relationship. If the measurement is proportional to the intensity of the wave, it will be a cosine squared relationship. Create a plot of relative intensity vs cosine and cosine squared to confirm how the device captures the information.

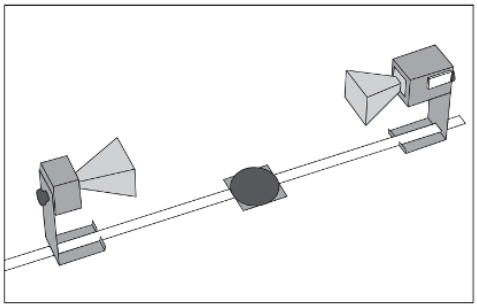


Figure 5.3: Microwave Apparatus Setup

Analysis

Report the measured index of refraction from your graph with its uncertainty. You have two values to report, one from the graph and the other from the critical angle measurement.
Provide the graph from the polarization experiment. Does this show the intensity or electric field?

Conclusion

Report your results from Procedure 1, 2 and 3. Make sure to compare your measured index of refraction with uncertainty to the accepted value of 1.49. For the dispersion experiment, what can you conclude about the relationship between index of refraction and wavelength?